Is LLM self-consistency a usable signal for routing annotation work? Project Proposal for NLP Course, Winter 2026/27

Tymoteusz Urban
Album number
email@pw.edu.pl

Sebastian Trojan
320664
01171271@pw.edu.pl

Mateusz Nizwantowski
313839
01161932@pw.edu.pl

supervisor: Soveatin Kuntur
Warsaw University of Technology
soveatin.kuntur.dokt@pw.edu.pl

Abstract

Write about the proposed work touching the research gap you are addressing, novelty, and on the proposed deliverable aimed from this project. (Short description up to 250 words.)

1 Introduction

1. **Scientific goal of the project.** This study asks whether self-consistency across LLM reruns predicts correctness of multilabel annotations of Polish persuasion and manipulation techniques and supports cost-effective human–model routing. H1: higher self-consistency distinguishes correct from incorrect predictions better than chance across models, datasets and prompting strategies. H2: at comparable annotation quality, consistency-based routing achieves greater automatic coverage than alternative confidence-based approaches, including cheaper single-run methods. We also evaluate risk–coverage trade-offs, human annotation effort saved and additional inference costs. Negative results are considered valid research outcomes.

2. **Significance of the project.** Annotation is costly, but LLMs can be consistently wrong. Repeated inference also adds compute cost, so improved routing must be compared with cheaper confidence signals. We evaluate self-consistency for multilabel Polish text classification and its practical cost–quality trade-off. Model comparisons are descriptive, not causal evidence about size or language specialisation.

3. **Concept and work plan.** Figure 1 summarises the workflow.

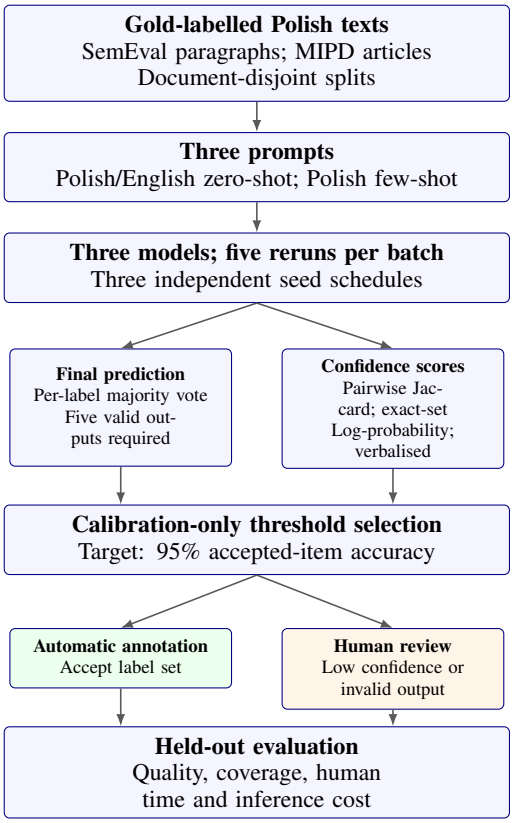


Figure 1: Planned annotation-routing pipeline. Thresholds are calibrated without test labels; human correction is simulated with gold labels.

Three deliverables. (i) Consistency–correctness analyses per model, dataset and supported label, with uncertainty estimates and confidence baselines. (ii) Risk–coverage and cost–quality curves, including comparison with single-run routing and a selected operating point (if feasible). (iii) Reproducible prompts, code, settings and outputs; sensitive outputs remain private.

Preliminary status. Datasets and label definitions are available; model experiments have not yet been run.

Timeline and ownership. Table 1 presents

weekly tasks, milestones and individual responsibilities.

Date	Planned output and owner
04 Nov	**Phase 1:** Methodology and pipeline (S); SOTA (T); datasets and data statement (M).
11 Nov	Docker/Conda/vLLM, GPU setup and data splits (M); inference pipeline and parsing (S); prompt preparation (T).
18 Nov	**Mid-term:** Three-model pilot with five reruns (S,M); SOTA presentation and video (T); start main runs (M).
25 Nov	Continue main inference (M); consistency metrics and confidence baselines (S); EDA and annotation-time sample (T).
02 Dec	**Phase 2:** Main inference and reproducible MR/README (M); preliminary evaluation (S); report and rebuttals (T).
09 Dec	Validate outputs and recover failures (M); AUC and confidence comparisons (S); preliminary error analysis (T).
16 Dec	Routing, calibration and bootstrap (S); GPU performance measurements (M); 20+ classified errors (T).
23 Dec	Risk–coverage and cost analysis (S); results discussion (T); reproducibility tests (M).
30 Dec	Complete report draft (T); final figures and results (S); code and README audit (M).
06 Jan	Final results verification (S); report revision and presentation video (T); final merge request (M).
13 Jan	**Phase 3:** Final report, presentation, code and MR (S,T,M).

Table 1: Weekly plan for 2026/27. S = Sebastian Trojan; T = Tymoteusz Urban; M = Mateusz Nizwantowski. All dates are Wednesdays; January dates refer to 2027.

Risks and responses. Pilot runtime will determine a feasible shared sample and GPU settings. Audit data/model licences, missing labels and document leakage. Report rare-label limitations; route invalid outputs to humans. Record uncertainty and examine at least 20 misclassified examples grouped by error type.

4. **Approach and research methodology.** *Data and models.* Use SemEval-2023 Task 3 Polish paragraphs and MIPD articles with separate native label taxonomies. Verify whether absent MIPD indicators mean negative or missing labels. Audit duplicates and enforce document-disjoint training, calibration and held-out splits; use official splits where possible. Models: Bielik 11B v3 Instruct (SpeakLeash), PLLuM 12B Chat 2412 (CYFRAGOVPL) and Gemma 4 E2B IT (Google).

Prompt strategies. Use Polish zero-shot, English zero-shot and Polish few-shot prompts with equivalent instructions, label IDs and output schemas. Few-shot examples come only from training data; comparisons isolate instruction language or demonstrations, not a full factorial design.

Repeated inference and failures. Generate five outputs per model, item and prompt using fixed stochastic settings and logged seeds. Replicate the main comparison using three independent five-output batches on a common pilot-sized sample, reporting the spread. Log raw and parsed outputs, versions and settings. Refer items with any invalid output to a human; report failures separately from classification accuracy and retain all items in coverage denominators.

Prediction and self-consistency. For five valid outputs, predict each label by majority vote. Measure exact-set agreement (modal-set frequency), mean pairwise Jaccard similarity over ten pairs (primary score), and per-label agreement. Confidence scores never use gold labels.

Confidence baselines. On the same voted predictions, compare self-consistency with length-normalised token log-probability of the canonical voted label set, scored by teacher forcing under the original prompt; verbalised confidence elicited on a fixed $[0, 1]$ scale; and random routing. Log additional scoring/query costs. Report unavailable log-probability baselines explicitly.

Metrics and uncertainty. Report micro/macro-F1 and exact-set accuracy on valid predictions, plus failure rates. Evaluate ROC AUC for predicting exact-set correctness, supplemented by error-detection PR AUC. Analyse per-label correctness only with adequate support. Use paired document-level bootstrap 95% confidence intervals and report spread across independent seed schedules.

Routing evaluation. Calibrate thresholds without test data, maximising coverage subject to 95% accepted-item exact-set accuracy

and a predeclared minimum accepted count; also examine 80% and 90% targets. Evaluate risk–coverage curves, accepted-only quality and final quality under ideal gold-based human correction on held-out data. Report infeasible targets instead of assuming coverage.

Cost–benefit analysis. Compare confidence signals on matched five-run predictions and complete policies against cheaper single-run baselines. Report coverage, quality, inference time and estimated human hours based on a timed sample from each dataset. Estimate net monetary cost using explicit labour and compute rates, including reruns and confidence queries; do not assume savings.

Tools. Use Python and vLLM on local NVIDIA GPUs, with a Dockerfile and Conda environment specification. Log versions, model settings, seeds, GPU memory, runtime and execution commands. No model fine-tuning is planned.

5. well-justified project literature (full bibliographic data);

6. **Data statement.** Source, licence, personal data and safeguards (up to six lines).

7. **Contribution table.** Table 2 records Phase 1 work.

Member	Phase 1 contribution	h
Tymoteusz Urban	State-of-the-art literature review.	8
Sebastian Trojan	Scientific goal, concept, methodology, plan, contribution table and LLM-use declaration.	5
Mateusz Nizwantowski	Datasets and status; data statement.	8
Total		17

Table 2: Phase 1 division of work.

8. **Declaration of LLM use.** OpenAI Codex supported drafting, language editing and LaTeX formatting. The authors are responsible for the final content.